

GLM-5.1 - Overview - Z.AI DEVELOPER DOCUMENT

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Overview

GLM-5.1 is Z.AI's latest flagship model, designed for long-horizon tasks. It can work continuously and autonomously on a single task for up to 8 hours, completing the full loop from planning and execution to iterative optimization and delivering production-grade results.

In both general capability and coding performance, GLM-5.1 is overall aligned with Claude Opus 4.6. It demonstrates stronger sustained execution in long-horizon autonomous tasks, complex engineering optimization, and real-world development workflows, making it an ideal foundation for building autonomous agents and long-horizon coding agents.

Positioning

Flagship Foundation Model

Input Modalities

Text

Output Modalities

Text

Context Length

200K

Maximum Output Tokens

128K

Capability

Thinking Mode

Offering multiple thinking modes for different scenarios

Streaming Output

Support real-time streaming responses to enhance user interaction experience

Function Call

Powerful tool invocation capabilities, enabling integration with various external toolsets

Context Caching

Intelligent caching mechanism to optimize performance in long conversations

Structured Output

Support for structured output formats like JSON, facilitating system integration

MCP

Flexibly integrate external MCP tools and data sources to expand application scenarios

Usage

Agentic Coding

General Conversation

Creative Writing

Artifacts / Front-End Development

Office Productivity

Introducing GLM-5.1

1

General and Coding Capability: Aligned with the Global Frontier

GLM-5.1 ranks among the world's top-tier models in both overall capability and coding performance, with overall performance aligned with Claude Opus 4.6 and leading results across multiple key benchmarks.

On SWE-Bench Pro, GLM-5.1 achieves a score of 58.4, outperforming GPT-5.4, Claude Opus 4.6, and Gemini 3.1 Pro, setting a new state-of-the-art result. At the same time, across 12 representative benchmarks covering reasoning, coding, agents, tool use, and browsing, GLM-5.1 demonstrates a broad and well-balanced capability profile.

This shows that GLM-5.1 is not a single-metric improvement. Instead, it advances simultaneously across general intelligence, real-world coding, and complex task execution, making it a stronger foundation for general-purpose agent systems and engineering production scenarios.

2

Long-Horizon Task Capability: Toward 8-Hour Sustained Execution

GLM-5.1 shows especially strong gains on long-horizon tasks, with major improvements in sustained execution, closed-loop optimization, and engineering delivery under complex objectives. Compared with models primarily designed for minute-level interactions, GLM-5.1 can work autonomously on a single task for up to 8 hours, completing the full process from planning and execution to testing, fixing, and delivery.

Under the same evaluation standard, GLM-5.1 is one of the few models capable of 8-hour sustained execution, and the first Chinese model to reach this level. The way we evaluate model capability is shifting from "how smart it is in a single turn" to "how long it can work reliably on a long-horizon task, and what it can actually deliver."

This capability is not simply about having a longer context window. It requires the model to maintain goal alignment over extended execution, reducing strategy drift, error accumulation, and ineffective trial and error, and enabling truly autonomous execution for complex engineering tasks.

Engineering Delivery: From Code Generation Toward Autonomous Agent

One of GLM-5.1's key breakthroughs is its ability to form an autonomous "experiment-analyze-optimize" loop in long-horizon tasks, rather than stopping at one-shot code generation. The model can proactively run benchmarks, identify bottlenecks, adjust strategies, and continuously improve results through iterative refinement.

In representative cases, GLM-5.1 can build a complete Linux desktop system from scratch within 8 hours. It can autonomously carry out 655 iterations, completing the entire optimization pipeline and boosting vector database query throughput to 6.9× that of the initial production version. On the KernelBench Level 3 optimization benchmark, it performs thousands of tool-invocation-driven optimizations on real machine learning workloads, achieving a 3.6× geometric mean speedup—significantly surpassing the 1.49× achieved by torch.compile in max-autotune mode.

These results show that GLM-5.1 is already capable of autonomous exploration, continuous improvement, and stable delivery in complex engineering environments, enabling it to take on higher-value tasks such as system building, performance optimization, and long-horizon coding agents.

Resources

API Documentation: Learn how to call the API.

Quick Start

The following is a full sample code to help you onboard GLM-5.1 with ease.

cURL

Official Python SDK

Official Java SDK

OpenAI Python SDK

Basic Call

```
curl -X POST "https://api.z.ai/api/paas/v4/chat/completions" \
  -H "Content-Type: application/json" \
  -H "Authorization: Bearer your-api-key" \
  -d '{
    "model": "glm-5.1",
    "messages": [
      {
        "role": "system",
        "content": "You are a senior full-stack software engineer, proficient in
frontend development, backend architecture design, and modern web technology stacks."
      },
      {
        "role": "user",
        "content": "Design and build a personal blog website for me, including a
homepage, article list page, and article detail page, using React + Node.js technology
stack."
      }
    ],
    "thinking": {
      "type": "enabled"
    },
    "max_tokens": 4096,
    "temperature": 1.0
  }'
```

Streaming Call

```
curl -X POST "https://api.z.ai/api/paas/v4/chat/completions" \
```

```
-H "Content-Type: application/json" \  
-H "Authorization: Bearer your-api-key" \  
-d '{  
  "model": "glm-5.1",  
  "messages": [  
    {  
      "role": "system",  
      "content": "You are a senior full-stack software engineer, proficient in  
frontend development, backend architecture design, and modern web technology stacks."  
    },  
    {  
      "role": "user",  
      "content": "Design and build a personal blog website for me, including a  
homepage, article list page, and article detail page, using React + Node.js technology  
stack."  
    }  
  ],  
  "thinking": {  
    "type": "enabled"  
  },  
  "stream": true,  
  "max_tokens": 4096,  
  "temperature": 1.0  
}'
```

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No

GLM-5.2

GLM-5

x

github

discord

linkedin